Table 1.1 A recap of the available goods, with their weight and calories.

Food	Weight (kgs)	Total calories
Potatoes	800	1,502,000
Wheat Flour	400	1,444,000
Rice	300	1,122,000
Beans (can)	300	690,000
Tomatoes (can)	300	237,000
Strawberry jam	50	130,000
Peanut butter	20	117,800

So, since the actual content is irrelevant for the decision (despite your understandable protests, mission control is very firm on that point), the only things that matters are the weight and total calories provided, for each of the boxes.

Hence, our problem can be abstracted as “choose any number of items from a set, **without the chance to take a fraction of any item**, such that their total weight won’t be over 1000 kgs, and in such a way that the total amount of calories is maximized”.

1.4.2 Looking for Solutions

Now that we have stated our problem, we can start looking for solutions.

You might be tempted to start packing your crate with boxes starting from the one with the highest calories value. That would be the potatoes box weighting 800 kgs.